Robust Estimation of Graph Statistics

E746C

1 Introduction

1.1 Background

1.2 Structure of this paper

1.3 Notations

Definition 1.1. *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community stochastic block model graph with $V = [n]$ and uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$. For each pair of vertices $i, j \in [n]$, we have*

$$\mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2}) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2}) & \text{if } x_i \neq x_j. \end{cases}$$

Definition 1.2. *The set of positive semidefinite matrices with diagonal entries equal to 1 is denoted by*

$$\mathcal{PSD}_1(n) = \{\boldsymbol{X} \in \mathbb{R}^{n \times n} : \boldsymbol{X} \succeq 0, \boldsymbol{X}_{ii} = 1 \text{ for all } i \in [n]\}.$$

Definition 1.3. *Given a matrix $\boldsymbol{M} \in \mathbb{R}^{n \times n}$, we define $\mathcal{SDP}(\boldsymbol{M}) := \max_{\boldsymbol{X} \in \mathcal{PSD}_1(n)} \langle \boldsymbol{M}, \boldsymbol{X} \rangle$.*

Definition 1.4. *Given a graph G, let $\boldsymbol{A}_G$ be its adjacency matrix, and define $\boldsymbol{A}_G^{\text{cent}} = \boldsymbol{A}_G - \mathbb{E}[\boldsymbol{A}_G]$. For $G \sim \mathcal{G}(n, d, \epsilon)$, we have $\mathbb{E}[\boldsymbol{A}_G] = \frac{d}{n}\mathbf{1}\mathbf{1}^\top$.*

2 (Looking at MS16)

introduce MS16

2.1 Algorithm

Algorithm 1 (SDP Algorithm from [MS16])**.**

1. Sample edges of G independently with probability $(1 + n^{-1/2})^{-1}$ to obtain the sampled edge set $E_{\text{sample}} \subset E$, and then compute the evaluation edge set $E_{\text{eval}} = E \setminus E_{\text{sample}}$.
2. Construct the subgraphs $G_{\text{sample}} = (V, E_{\text{sample}})$ and $G_{\text{eval}} = (V, E_{\text{eval}})$.
3. Compute an optimiser $\boldsymbol{X}^*$ of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$, i.e., $\langle \boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle = \mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$.
4. Compute the eigenvectors $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$ of $\boldsymbol{X}^*$.
5. For each $i, j \in [n]$, define $\hat{\boldsymbol{x}}^{(i,j)} \in \{-1, 0, 1\}^n$ as follows:

$$\hat{\boldsymbol{x}}_k^{(i,j)} = \begin{cases} \operatorname{sign}(\boldsymbol{v}_k^{(i)}) & \text{if } |\boldsymbol{v}_k^{(i)}| \geq |\boldsymbol{v}_j^{(i)}|, \\ 0 & \text{otherwise.} \end{cases}$$

6. Find $(i^*, j^*) = \arg\max_{i,j \in [n]} (\hat{\boldsymbol{x}}^{(i,j)})^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)} = \sum_{a,b \in E_{\text{eval}}} \hat{\boldsymbol{x}}_a^{(i,j)} \cdot \hat{\boldsymbol{x}}_b^{(i,j)}$.

7. Output $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G) = \hat{\boldsymbol{x}}^{(i^*,j^*)}$.

We have the following weak recovery rate guarantee for Algorithm 1.

Theorem 2.1 (Weak Recovery for Algorithm 1)**.** *Let $G \sim \mathcal{G}(n, d, \epsilon)$ and assume, for some $\delta > 0$, that $\lambda = \epsilon\sqrt{d}/2 \geq 1 + \delta$. Then, there exists $\eta = \eta(\delta) > 0$ and $d_* = d_*(\delta) > 0$ such that, for all $d \geq d_*$, the output of Algorithm 1, $\hat{\boldsymbol{x}}^{\mathrm{SDP}}(G)$, satisfies*

$$\mathbb{P}\left(\frac{1}{n}|\langle \hat{\boldsymbol{x}}^{\mathrm{SDP}}(G), x\rangle| \geq \eta\right) \geq 1 - Ce^{-\sqrt{n}/C}.$$

discuss robustness of the algorithm

3 Proof of Weak Recovery (Theorem 2.1)

restructure and move main theorem to this section

Before we start the proof, we will make a few definitions, and show a few preliminary results. Let $\boldsymbol{x}$ be the label vector of $G \sim \mathcal{G}(n, d, \epsilon)$. Note that $G_{\mathrm{sample}} \sim \mathcal{G}(n, d_{\mathrm{sample}}, \epsilon)$, and $G_{\mathrm{eval}} \sim \mathcal{G}(n, d_{\mathrm{eval}}, \epsilon)$, where $d_{\mathrm{sample}} = d/(1 + n^{-1/2})$ and $d_{\mathrm{eval}} = n^{-1/2} d_{\mathrm{sample}}$. For simplicity, from now on we write $d = d_{\mathrm{sample}}$.

Throughout the proof, we will use the constant C to denote a positive constant that changes from line to line. We will often use the fact that, if two events both hold with probability at least $1 - Ce^{-n/C}$, then their intersection also holds with probability at least $1 - Ce^{-n/C}$, by a simple union bound, and

$$C_1 e^{-n/C_1} + C_2 e^{-n/C_2} \leq (C_1 + C_2) e^{-n/\max(C_1, C_2)} \leq (C_1 + C_2) e^{-n/(C_1+C_2)} = Ce^{-n/C}.$$

Also, note that, in our definition, G_{sample} and G_{eval} are dependent. However, we need their independence in our proof. To fix this issue, we note that if we sampled them independently, then the two graphs

investigate independence

3.1 Step 1: $\boldsymbol{X}^*$ contains nontrivial information about $\boldsymbol{x}$

We first introduce a group of Gaussian random matrices (GOEs) that have well-behaved SDP values.

Definition 3.1 (Gaussian orthogonal ensemble (GOE) and the deformed GOE $\boldsymbol{B}(\lambda)$)**.** *A random matrix $\boldsymbol{W} \in \mathbb{R}^{n\times n}$ is a* GOE matrix *if it is a symmetric matrix with $\{\boldsymbol{W}_{ij}\}_{i\leq j}$ independent and $\boldsymbol{W}_{ij} \sim \mathcal{N}(0, 1/n)$ for $i < j$ and $\boldsymbol{W}_{ii} \sim \mathcal{N}(0, 2/n)$. The* deformed GOE matrix $\boldsymbol{B}(\lambda)$ *is then defined as*

$$\boldsymbol{B}(\lambda) = \frac{\lambda}{n}\mathbf{1}\mathbf{1}^\top + \boldsymbol{W}.$$

We then have the following phase transition result for the SDP value of $\boldsymbol{B}(\lambda)$.

Lemma 3.1 (SDP value of $\boldsymbol{B}(\lambda)$, Theorem 5 of [MS16])**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix defined in Definition 3.1. Then, with probability converging to 1 as $n \to \infty$,*

(a) if $\lambda \leq 1$: for any $\epsilon > 0$, we have $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \in [2 - \epsilon, 2 + \epsilon]$.

(b) if $\lambda > 1$: there exists $\Delta(\lambda) > 0$ such that $\mathcal{SDP}(\boldsymbol{B}(\lambda))/n \geq 2 + \Delta(\lambda)$.

If we set the diagonal entries of $\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}$ to be $\lambda\sqrt{d}/n$, then

$$\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{E},$$

where $\boldsymbol{E}$ is a symmetric matrix with zero diagonal and independent off-diagonal entries satisfying

$$\boldsymbol{E}_{ij} = \begin{cases} \frac{1}{\sqrt{d}} - \frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } \mathbb{P}((i,j) \in E_{\text{sample}}), \\ -\frac{\sqrt{d}}{n} - \frac{\lambda}{n}\boldsymbol{x}_i\boldsymbol{x}_j & \text{with probability } 1 - \mathbb{P}((i,j) \in E_{\text{sample}}). \end{cases}$$

Let $\boldsymbol{X}^*$ be an optimizer of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}})$. We will now try to bound $\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle$ and $\langle \frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \rangle$ away from each other using the phase transition result of Lemma 3.1, which will then give us a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$, which is the correlation between $\boldsymbol{X}^*$ and the ground truth labels matrix $\boldsymbol{x}\boldsymbol{x}^\top$.

The following lemma bounds $\mathcal{SDP}(\boldsymbol{E})$ to the SDP value of a GOE matrix.

Lemma 3.2 (GOE approximation of $\mathcal{SDP}(\boldsymbol{E})$)**.** *Let $\boldsymbol{W}$ be a GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$ with probability at least $1 - Ce^{-n/C}$, we have*

$$\left| \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We also have a similar result bounding $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$ by the SDP value of a deformed GOE matrix.

Lemma 3.3 (Deformed GOE approximation of $\mathcal{SDP}(\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}/\sqrt{d})$)**.** *Let $\boldsymbol{B}(\lambda)$ be a deformed GOE matrix. Then, there exists $C = C(\lambda) > 0$ such that, for all $n \geq n_0(d, \epsilon)$*

$$\left| \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) \right| \leq \frac{C\log(d)}{d^{1/10}}.$$

Moreover, $C(\lambda)$ is bounded for λ in any compact set.

We will discuss the proofs of the above two lemmas later in **??**. We will take them for granted here.

Now we have the tools to obtain a lower bound on $n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \rangle$.

$$\begin{aligned}
\left\langle \frac{\lambda}{n^2}\boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^* \right\rangle &= \frac{1}{n}\left(\left\langle \frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}, \boldsymbol{X}^* \right\rangle - \langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \right) \\
&= \frac{1}{n}\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}\right) - \frac{1}{n}\langle \boldsymbol{E}, \boldsymbol{X}^* \rangle \\
&\geq \frac{1}{n}\mathcal{SDP}(\frac{1}{\sqrt{d}}\boldsymbol{A}^{\text{cent}}_{G_{\text{sample}}}) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{E}) && \text{by optimality of SDP} \\
&\geq \frac{1}{n}\mathcal{SDP}(\boldsymbol{B}(\lambda)) - \frac{1}{n}\mathcal{SDP}(\boldsymbol{W}) - \frac{C\log(d)}{d^{1/10}} && \text{by Lemmas 3.2 and 3.3} \\
&\geq \Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}}, && \text{by Lemma 3.1}
\end{aligned}$$

with probability at least $1 - Cn^{-4}$ (see the remark below).

Remark 3.1. In the original paper [MS16], the authors claimed that the above inequality holds with probability at least $1 - Ce^{-n/C}$, but tracking the proof of Lemma 3.1 (Appendices F and G of [MS16]) shows that they only proved the result with probability at least $1 - Cn^{-4}$ (Lemma G.8 of [MS16]). This is still sufficient for our purposes, as the probability still tends to 1 as $n \to \infty$.

Now fix λ, and take d sufficiently large such that $\Delta_1(\lambda) - \frac{C\log(d)}{d^{1/10}} \geq \lambda\Delta_2(\lambda) > 0$. Then, we have, with probability at least $1 - Cn^{-4}$,

$$n^{-2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle \geq \Delta_2(\lambda).$$

This is an important milestone, as it shows that $\boldsymbol{X}^*$ has a nontrivial correlation with the ground truth $\boldsymbol{x}\boldsymbol{x}^\top$.

3.2 Step 2: There exists "good" label estimators among $\{\hat{\boldsymbol{x}}^{(i,j)}\}_{i,j\in[n]}$

We first define the following rounding function used in step 5 of Algorithm 1:

Definition 3.2 (Rounding function)**.** *For $t \geq 0$, define the* rounding function $r_t : \mathbb{R} \to \{-1, 0, +1\}$ *as follows:*

$$r_t(x) = \begin{cases} 1 & \text{if } x \geq t, \\ 0 & \text{if } -t < x < t, \\ -1 & \text{if } x \leq -t. \end{cases}$$

Hence, we can write $\hat{\boldsymbol{x}}^{(\hat{\imath},j)} = r_{\boldsymbol{v}_j^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})})$.

Since $\boldsymbol{X} \in \mathcal{PSD}_1(n)$, writing its eigen-decomposition as $\boldsymbol{X} = \sum_{i=1}^n \xi_i \boldsymbol{v}^{(i)}(\boldsymbol{v}^{(i)})^\top$, where ξ_i is the i-th largest eigenvalue, and $\boldsymbol{v}^{(i)}$ is the corresponding unit length eigenvector, we have $\xi_i \geq 0$ for all i by positive semidefiniteness, and $\sum_{i=1}^n \xi_i = \mathrm{Tr}(\boldsymbol{X}) = n$. Thus,

$$\Delta_2(\lambda) \leq \frac{1}{n^2}\langle \boldsymbol{x}\boldsymbol{x}^\top, \boldsymbol{X}^*\rangle = \frac{1}{n^2}\sum_{i=1}^n \xi_i(\boldsymbol{x}^\top\boldsymbol{v}^{(i)})^2 = \sum_{i=1}^n \frac{\xi_i}{n}\cdot\left(\frac{\boldsymbol{x}^\top\boldsymbol{v}^{(i)}}{\sqrt{n}}\right)^2.$$

So there exists $\hat{\imath} \in [n]$ such that $|\boldsymbol{x}^\top\boldsymbol{v}^{(\hat{\imath})}| \geq \sqrt{n\Delta_2(\lambda)}$.

Next, we will use following lemma to justify that the rounding function preserves this correlation, which shows that it is possible to extract a good label estimator.

Lemma 3.4 (Rounding Lemma)**.** *Let X, Y be random variables such that: $\mathbb{P}(X = +1) = \mathbb{P}(X = -1) = 1/2$, $\mathbb{E}(XY) \geq \epsilon > 0$, and $\mathbb{E}(Y^2) = 1$. Then, there exists t^* depending on the joint distribution of X and Y, such that*

$$\mathbb{E}(Xr_{t^*}(Y)) \geq \frac{\epsilon^2}{2}.$$

Proof. Note that for all t, we have $Xr_t(Y) = r_t(XY)$, and $Y^2 = (XY)^2$. Let $Z = XY$, then it suffices to show that given $\mathbb{E}(Z) \geq \epsilon > 0$ and $\mathbb{E}(Z^2) = 1$, we have $\mathbb{E}(r_t(Z)) \geq \epsilon^2/2$. Recall that for nonnegative

random variable S, by Fubini, we have

$$\mathbb{E}(S) = \mathbb{E}\left(\int_0^\infty \mathbb{1}_{s\le S} ds\right) = \int_0^\infty \mathbb{P}(S \ge s) ds, \quad \mathbb{E}(S^2) = \mathbb{E}\left(\int_0^\infty 2s\mathbb{1}_{s\le S} ds\right) = \int_0^\infty 2s\mathbb{P}(S \ge s) ds.$$

Therefore, writing $Z^\pm = \max(\pm Z, 0)$, we have

$$\begin{aligned}
\epsilon \le \mathbb{E}(Z) = \mathbb{E}(Z_+) - \mathbb{E}(Z_-) &= \int_0^\infty \mathbb{P}(Z_+ \ge z) dz - \int_0^\infty \mathbb{P}(Z_- \ge z) dz \\
&\le \int_0^K (\mathbb{P}(Z \ge z) - \mathbb{P}(Z \le -z)) dz + \int_K^\infty \frac{2z}{2K}\mathbb{P}(Z \ge z) dz - \int_K^\infty \frac{2z}{2K}\mathbb{P}(-Z \ge z) dz \\
&= \int_0^K \mathbb{E} r_t(Z) dt + \frac{1}{2K}\mathbb{E}(Z_+^2 - Z_-^2) = \int_0^K \mathbb{E} r_t(Z) dt + \frac{1}{2K}.
\end{aligned}$$

Now take $K = 1/\epsilon$, then

$$\frac{1}{K}\int_0^K \mathbb{E} r_t(Z) dt \ge \frac{1}{K}(\epsilon - \frac{1}{2K}) = \frac{\epsilon^2}{2}$$

So the average of $\mathbb{E} r_t(Z)$ over $t \in [0, 1/\epsilon]$ is greater than $\epsilon^2/2$, so there exists $t^* \in [0, 1/\epsilon]$ such that $\mathbb{E} r_{t^*}(Z) \ge \epsilon^2/2$. □

Without loss of generality, assume that $\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} > 0$. Consider (X, Y) drawn uniformly from $\{(\boldsymbol{x}_k, \sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})})\}_{k\in[n]}$, then we have $\mathbb{E}(XY) = n^{-1/2}\boldsymbol{x}^\top \boldsymbol{v}^{(\hat{\imath})} \ge \sqrt{\Delta_2(\lambda)}$ and $\mathbb{E}(Y^2) = n^{-1}||\sqrt{n}\boldsymbol{v}^{(\hat{\imath})}||^2 = ||\boldsymbol{v}^{(\hat{\imath})}||^2 = 1$. Thus, by Lemma 3.4, there exists $t^* \in [0, 1/\sqrt{\Delta_2(\lambda)}]$ such that

$$\frac{\Delta_2(\lambda)}{2} \le \mathbb{E}(X r_{t^*}(Y)) = \frac{1}{n}\sum_{k=1}^n \boldsymbol{x}_k r_{t^*}(\sqrt{n}\boldsymbol{v}_k^{(\hat{\imath})}) = \frac{1}{n}\langle \boldsymbol{x}, r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})})\rangle,$$

where $\tilde{t} = t^*/\sqrt{n} \in [0, 1/\sqrt{n\Delta_2(\lambda)}]$.

Remark 3.2. Here, we slightly improved the rounding lemma bound from $\epsilon^2/4$ in [MS16] to $\epsilon^2/2$, although the main proof idea remains the same. Also, the original application of the rounding lemma missed the scaling factor $\sqrt{n}$ in the definition of Y, but the conclusion remains valid.

Finally, note that for any value of $\tilde{t}$, we can find $\hat{\jmath} \in [n]$ such that $r_{\tilde{t}}(\boldsymbol{v}^{(\hat{\imath})}) = r_{\boldsymbol{v}_{\hat{\jmath}}^{(\hat{\imath})}}(\boldsymbol{v}^{(\hat{\imath})}) = \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}$. Therefore, there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})}\rangle \ge \frac{\Delta_2(\lambda)}{2}. \tag{3.1}$$

Now, it remains to show that the selection step (Step 6) of Algorithm 1 will find such $\hat{\imath}, \hat{\jmath}$ with high probability, which will then complete the proof of Theorem 2.1.

3.3 Step 3: A "good" estimator can be found with G_{eval}

We first want to relate the quantities $\hat{\boldsymbol{x}}^\top \boldsymbol{A}_{G_{\text{eval}}}^{\text{cent}} \hat{\boldsymbol{x}}$, the objective function in Step 6 of Algorithm 1, to $\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle$, the correlation above, with high probability. Recall the well-known Azuma-Hoeffding inequality. add reference

Lemma 3.5 (Azuma-Hoeffding inequality)**.** *Let $X_0, \ldots, X_m$ be a martingale, with bounded increments, i.e. $|X_i - X_{i-1}| \le c_i$ for all $i \in [m]$. Then, for all $t > 0$, we have*

$$\mathbb{P}(|X_m - \mathbb{E}(X_m)| \ge t) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^m c_i^2}\right).$$

We apply this inequality to the objective function in Step 6 of Algorithm 1, and obtain the following concentration result.

Lemma 3.6. *Let $G \sim \mathcal{G}(n, d, \epsilon)$. For any fixed vector $\hat{\boldsymbol{x}} \in \{-1, 0, +1\}^n$, define $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}}$. Then, for any $t > 0$,*

$$\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E_{\text{eval}}| \le m\Big) \le 2\exp\left(-\frac{t^2}{8m}\right).$$

Proof. Note that $f(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G^{\text{cent}} \hat{\boldsymbol{x}} = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}} - (d/n)(\hat{\boldsymbol{x}}^\top \mathbf{1})^2$, so writing $\tilde{f}(G; \hat{\boldsymbol{x}}) = \hat{\boldsymbol{x}}^\top \boldsymbol{A}_G \hat{\boldsymbol{x}}$, we have $f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}})) = \tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))$. Note that $\tilde{f}(G; \hat{\boldsymbol{x}}) = \sum_{(a,b) \in E} 2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b$.

For some ordering of all possible edges, we can define a filtration $\mathcal{F}_0, \mathcal{F}_1, \ldots, \mathcal{F}_m$ with $\mathcal{F}_k = \sigma(E_1, \ldots, E_k)$, where $E_k \in E(G) \cup \varnothing$ is the k-th edge of G in the ordering. Then, define the martingale $X_k = \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | \mathcal{F}_k)$ for $k \in [m]$. Since $|E(G)| \le m$, we have $X_m = \tilde{f}(G; \hat{\boldsymbol{x}})$. Note that for all $k \in [m]$, we have

$$\begin{aligned}
|X_k - X_{k-1}| &= \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1})\Big| \\
&\le \sup_{e_{1:k-1}, e_k, e_k'} \Big|\mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}) | E_{1:k-1} = e_{1:k-1}, E_k = e_k')\Big| \\
&\le \sup_{E(G^+) = E(G) \cup \{(a,b)\}} \Big|\tilde{f}(G^+; \hat{\boldsymbol{x}}) - \tilde{f}(G; \hat{\boldsymbol{x}})\Big| \\
&\le \sup_{a,b} |2\hat{\boldsymbol{x}}_a \hat{\boldsymbol{x}}_b| \le 2.
\end{aligned}$$

Hence, we can apply the Azuma-Hoeffding inequality Lemma 3.5 to X with $c_i = 2$ for all $i \in [m]$ to obtain

$$\begin{aligned}
\mathbb{P}\Big(\big|f(G; \hat{\boldsymbol{x}}) - \mathbb{E}(f(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E(G)| \le m\Big) &= \mathbb{P}\Big(\big|\tilde{f}(G; \hat{\boldsymbol{x}}) - \mathbb{E}(\tilde{f}(G; \hat{\boldsymbol{x}}))\big| \ge t \;\Big|\; |E(G)| \le m\Big) \\
&= \mathbb{P}(|X_m - \mathbb{E}(X_m)| \ge t) \\
&\le 2\exp\left(-\frac{t^2}{8m}\right).
\end{aligned}$$

as desired. □

To apply the above concentration result, we need to bound $|E_{\text{eval}}| \sim \text{Binomial}\left(|E|, n^{-1/2}/(1 + n^{-1/2})\right)$, which we can achieve with the following Chernoff bound argument.

Lemma 3.7 (Bounding $|E_{\text{eval}}|$)**.** *Let $p_n = n^{-1/2}/(1 + n^{-1/2})$. Then, for some constant $C > 0$, we have*

$$\mathbb{P}(|E_{\text{eval}}| \ge C\sqrt{n}) \le Ce^{-\sqrt{n}/C}.$$

Proof. Recall that $|E| \sim \mathrm{Binomial}\left(\binom{n}{2}, d/n\right)$, so by Chernoff bound, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E| \geq K_0 n) &\leq e^{-sK_0 n}\mathbb{E}(e^{s|E|}) = e^{-sK_0 n}\left(1 - \frac{d}{n} + \frac{d}{n}e^s\right)^{\binom{n}{2}} \\
&\leq \exp\left(-sK_0 n + \frac{d}{n}(e^s - 1)\binom{n}{2}\right) \\
&\leq \exp\left(\left(\frac{d}{2}(e^s - 1) - sK_0\right)n\right) \\
&\leq C_0 e^{-n/C_0},
\end{aligned}$$

where the last inequality holds when K_0 is sufficiently large, and C_0 is a constant depending on d.

Now, on the event $\{|E| \leq K_0 n\}$ with probability at least $1 - C_0 e^{-n/C_0}$, we have $|E_{\mathrm{eval}}| \sim \mathrm{Binomial}\left(|E|, p_n\right)$, so by Chernoff bound again, for all $s > 0$, we have

$$\begin{aligned}
\mathbb{P}(|E_{\mathrm{eval}}| \geq K_1\sqrt{n}) &= \mathbb{E}\left(\mathbb{P}(|E_{\mathrm{eval}}| \geq K_1\sqrt{n} \mid |E|)\right) \leq e^{-sK_1\sqrt{n}}\mathbb{E}(\mathbb{E}(e^{s|E_{\mathrm{eval}}|} \mid |E|)) \\
&= e^{-sK_1\sqrt{n}}\mathbb{E}((1 - p_n + p_n e^s)^{|E|}) \leq e^{-sK_1\sqrt{n}}\left(1 + p_n(e^s - 1)\right)^{K_0 n} \\
&= \exp\left(-sK_1\sqrt{n} + K_0(e^s - 1)np_n\right) \leq \exp\left((K_0(e^s - 1) - sK_1)\sqrt{n}\right) \\
&\leq C_1 e^{-\sqrt{n}/C_1},
\end{aligned}$$

where the last inequality holds when K_1 is sufficiently large, and C_1 is a constant depending on d. Combining the two probability bounds, we have

$$\mathbb{P}(|E_{\mathrm{eval}}| \geq K\sqrt{n}) \leq C_0 e^{-n/C_0} + C_1 e^{-\sqrt{n}/C_1} \leq C e^{-\sqrt{n}/C},$$

for suitable constants C, as desired. □

expand the following argument slightly

By Lemmas 3.6 and 3.7, for all $s \in [0, 1]$, we have, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$,

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}} - \mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}}\right)\right| \leq s\sqrt{n}.$$

Note that

$$\begin{aligned}
\mathbb{E}\left(\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}}\right) &= \hat{\boldsymbol{x}}^\top \mathbb{E}\left(\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\right)\hat{\boldsymbol{x}} \\
&= \hat{\boldsymbol{x}}^\top \frac{\epsilon d}{2n}\left(\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{I}_n\right)\hat{\boldsymbol{x}} \\
&= \frac{\epsilon d}{2n}(\hat{\boldsymbol{x}}^\top \boldsymbol{x})^2 - \frac{\epsilon d}{2}.
\end{aligned}$$

Thus, with probability at least $1 - Ce^{-\sqrt{n}s^2/C}$, we have

$$\left|\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}\hat{\boldsymbol{x}} - \frac{\epsilon d}{2n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}\rangle^2 + \frac{\epsilon d}{2}\right| \leq s\sqrt{n}. \tag{3.2}$$

This concentration result relates $\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}$ to the correlation $\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle$ with high probability.

In particular, at the end of the previous step (Equation (3.1)), we have shown that there exists some $\hat{\imath}, \hat{\jmath} \in [n]$ such that $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \rangle \geq \frac{\Delta_2(\lambda)}{2}$. Then, by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$ (taking $s = 1$), we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \rangle\right)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{8}\Delta_2(\lambda)^2 - \frac{\epsilon d}{2n} - \frac{1}{\sqrt{n}} \geq \frac{\epsilon d}{12}\Delta_2(\lambda)^2 =: \Delta_3(\lambda),$$

where the last inequality follows from

$$\frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{24}\Delta_2(\lambda)^2,$$

which holds for sufficiently large n. Therefore,

$$\max_{i,j\in[n]} \frac{1}{n}\hat{\boldsymbol{x}}^{(i,j)\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)} \geq \frac{1}{n}\hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(\hat{\imath},\hat{\jmath})} \geq \Delta_3(\lambda).$$

On the other hand, suppose that $\hat{\boldsymbol{x}}$ has a lower correlation with $\boldsymbol{x}$, say $\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle \leq \frac{\Delta_2(\lambda)}{4}$, then by the concentration result 3.2, with probability at least $1 - Ce^{-\sqrt{n}/C}$, we have

$$\frac{1}{n}\hat{\boldsymbol{x}}^\top \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}} \leq \frac{\epsilon d}{2}\left(\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}} \rangle\right)^2 + \frac{\epsilon d}{2n} + \frac{1}{\sqrt{n}} \leq \frac{\epsilon d}{32}\Delta_2(\lambda)^2 + \frac{\epsilon d}{24}\Delta_2(\lambda)^2 < \frac{\epsilon d}{12}\Delta_2(\lambda)^2 = \Delta_3(\lambda).$$

Thus, with probability at least $1 - Ce^{-\sqrt{n}/C}$, the maximizer $\hat{\boldsymbol{x}}^{\text{SDP}} = \hat{\boldsymbol{x}}^{(i^*,j^*)}$ of $\max_{i,j\in[n]} \hat{\boldsymbol{x}}^{(i,j)\top} \boldsymbol{A}^{\text{cent}}_{G_{\text{eval}}} \hat{\boldsymbol{x}}^{(i,j)}$ must satisfy

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{\text{SDP}} \rangle \geq \frac{\Delta_2(\lambda)}{4}.$$

This completes the proof of Theorem 2.1. ■

4 Robustness of the SDP

4.1 Robustness against modifying $O(n\sqrt{d})$ edges

explain

4.2 Robustness with perturbed SBMs

We first define the following more general class of random graph models, the *perturbed SBM*. We will then show that the same algorithm still works for this more general class of random graph models, which demonstrates the robustness of the algorithm and the SDP approach in general.

Definition 4.1 (Perturbed SBM). *A graph $G \sim \mathcal{G}(n, d, \epsilon)$ is a balanced, two-community h-perturbed stochastic block model graph with $V = [n]$, uniformly random label vector $\boldsymbol{x} \in \{\pm 1\}^n$ satisfying $\mathbf{1}^\top \boldsymbol{x} = 0$, i.i.d. latent variables $\xi_i \sim \text{Unif}([0,1])$ for $i \in [n]$, and perturbation function $h : [0,1]^2 \to \mathbb{R}$ satisfying*

$\mathbb{E}[h(\xi_i, \xi_j)] = 0$ *for all* $i, j \in [n]$*, if for each pair of vertices* $i, j \in [n]$*, we have*

$$p_{ij} := \mathbb{P}((i,j) \in E) = \begin{cases} \frac{d}{n}(1 + \frac{\epsilon}{2} + h(\xi_i, \xi_j)) & \text{if } x_i = x_j, \\ \frac{d}{n}(1 - \frac{\epsilon}{2} + h(\xi_i, \xi_j)) & \text{if } x_i \neq x_j. \end{cases}$$

We write $\eta_{ij} = x_i x_j \epsilon/2 + h(\xi_i, \xi_j)$*, and* $h_{ij} = h(\xi_i, \xi_j)$ *for short.*

We have the following weak recovery result for the perturbed SBM.

Theorem 4.1 (Weak recovery for perturbed SBM)**.** *Let* $G \sim \mathcal{G}(n, d, \epsilon)$ *be a balanced, two-community* h*-perturbed SBM graph, and let* $\lambda = \epsilon^2 d/4$*. Suppose there exists* $\alpha > 0$*, such that:*

(a) the perturbation h *satisfies that* $\|h\|_\infty \leq C d^{-2\alpha}$*,*

(b) let $\boldsymbol{H}$ *be the matrix with entries* $H_{ij} = h(\xi_i, \xi_j)$*, then* $\mathbb{E}(\|\boldsymbol{H}\|_{op}) \leq C n d^{-1/2-\alpha}$*.*

If $\lambda > 1$*, then there exists some constant* $\Delta_4(\lambda) > 0$*, such that with probability at least* $1 - Cn^{-4}$ *, the estimator* $\hat{\boldsymbol{x}}^{\mathrm{SDP}}$ *returned by Algorithm 1 satisfies*

check the probability bound

$$\frac{1}{n}\langle \boldsymbol{x}, \hat{\boldsymbol{x}}^{\mathrm{SDP}} \rangle \geq \Delta_4(\lambda).$$

Remark 4.1. We can view the theorem above as a different notion of robustness of the algorithm, where instead of directly editing edges, we are allowed to perturb the edge probabilities by some random $h(\xi_i, \xi_j)$. The theorem shows that as long as the perturbation is not too large or too structured, the algorithm can still achieve weak recovery.

4.3 Comparison of the two notions of robustness

Note that the perturbation can lead to more than $O(n\sqrt{d})$ edge modifications in expectation.

Roughly speaking, the operator norm bound on $\boldsymbol{H}$ in condition (b) of Theorem 4.1 controls the degree of structure in the perturbation. For example, if $h(\xi_i, \xi_j)$'s are i.i.d., then $\mathbb{E}\|\boldsymbol{H}\|_{\mathsf{op}} \sim O(\sqrt{n})$, and since $d \ll n$, this bound is easily satisfied, with individual edge perturbations of order $O(d^{-2\alpha})$ to satisfy the L_∞ bound in condition (a). On the other hand, if $h(\xi_i, \xi_j)$'s are highly structured, say $\boldsymbol{H} = \boldsymbol{u}\boldsymbol{u}^\top$ for some vector $\boldsymbol{u} \in \mathbb{R}^n$, then $\|\boldsymbol{H}\|_{\mathsf{op}} = O(n)$, so the individual edge perturbations must be of order $O(d^{-1/2-\alpha})$.

flesh out the above argument with more citations

concrete example with more than $O(n\sqrt{d})$ edge modifications

5 Proof of weak recovery for perturbed SBMs (Theorem 4.1)

Note that the proof of weak recovery in Section 3 only uses the adjacency matrix of the graph in two places: the SDP approximation lemmas (Lemmas 3.2 and 3.3), which we will discuss for the rest of this section, and $\mathbb{E}[\boldsymbol{A}^{\mathrm{cent}}_{G_{\mathrm{eval}}}]$ in Equation (3.2), which will be the same for the perturbed SBM and the original SBM since $\mathbb{E}[h(\xi_i, \xi_j)] = 0$ for all $i, j \in [n]$. Thus, to show that the algorithm remains effective for the perturbed SBM, we only need to show that similar SDP approximation results hold for the perturbed SBM.

Therefore, in this section, we will demonstrate how to get a similar SDP approximation result as in Lemmas 3.2 and 3.3, which will prove theorem 4.1. The proofs of the lemmas will follow the exact same route, so they are ommitted here for brevity. We refer the reader to the original paper [MS16] for more details.

5.1 Helper functions

The main idea of approximating the SDP of adjacency matrices with the SDP of deformed GOE matrices is to utilize the following two helper functions:

Definition 5.1 ($\mathcal{OPT}_k$)**.** *Define the rank-constrained SDP value for $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ and $k \in [n]$:*

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\substack{\boldsymbol{X}\in\mathcal{PSD}_1(n)\\ \mathrm{rank}(\boldsymbol{X})\le k}} \langle \boldsymbol{M}, \boldsymbol{X}\rangle.$$

Equivalently, let $\mathbb{S}^{k-1} = \{\boldsymbol{v} \in \mathbb{R}^k : ||\boldsymbol{v}||_2 = 1\}$ be the unit sphere in $\mathbb{R}^k$, then we have

$$\mathcal{OPT}_k(\boldsymbol{M}) = \max_{\boldsymbol{\sigma}_1,\ldots,\boldsymbol{\sigma}_n\in\mathbb{S}^{k-1}} \sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle.$$

Definition 5.2 (Smoothed OPT)**.** *For $\beta > 0$, define $\Phi(\beta, k; \boldsymbol{M})$, the smoothed version of $\mathcal{OPT}_k(\boldsymbol{M})$, as follows:*

$$\Phi(\beta, k; \boldsymbol{M}) = \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right),$$

where ν is the uniform probability measure on $(\mathbb{S}^{k-1})^n$.

We can see that as $\beta \to \infty$, $\Phi(\beta, k; \boldsymbol{M})$ will select the maximum value of $\sum_{i,j=1}^{n} M_{ij}\langle \boldsymbol{\sigma}_i, \boldsymbol{\sigma}_j\rangle$, which is exactly $\mathcal{OPT}_k(\boldsymbol{M})$.

In their paper, Montanari and Sen showed the following bounds relating $\mathcal{SDP}(\boldsymbol{M})$, $\mathcal{OPT}_k(\boldsymbol{M})$ and $\Phi(\beta, k; \boldsymbol{M})$. We will quote these results without proof.

Theorem 5.1 (SDP vs OPT, Theorem 4 Equation 15 of [MS16])**.** *For $k \ge k_0$, where k_0 is a sufficiently large constant, for any symmetric matrix $\boldsymbol{M} \in \mathbb{R}^{n\times n}$, we have*

$$\left(1 - \frac{1}{k}\right)\mathcal{SDP}(\boldsymbol{M}) - \frac{1}{k}|\mathcal{SDP}(-\boldsymbol{M})| \le \mathcal{OPT}_k(\boldsymbol{M}) \le \mathcal{SDP}(\boldsymbol{M}).$$

Remark 5.1. Using just the first inequality, we have

$$\mathcal{OPT}_k(\boldsymbol{M}) + \mathcal{OPT}_k(-\boldsymbol{M}) \ge \left(1 - \frac{2}{k}\right)\left(|\mathcal{SDP}(\boldsymbol{M})| + |\mathcal{SDP}(-\boldsymbol{M})|\right),$$

so subtract by $\mathcal{SDP}(\boldsymbol{M})$ on every term and flipping signs, we get

$$0 \leq \mathcal{SDP}(\boldsymbol{M}) - \mathcal{OPT}_k(\boldsymbol{M}) \leq \frac{1}{k}\left(|\mathcal{SDP}(\boldsymbol{M})| + |\mathcal{SDP}(-\boldsymbol{M})|\right) \leq \frac{1}{k-2}\left(\mathcal{OPT}_k(\boldsymbol{M}) + \mathcal{OPT}_k(-\boldsymbol{M})\right).$$

Remark 5.2. Clearly, $\boldsymbol{A}_G^{\text{cen}}$ and $\boldsymbol{B}(\lambda)$ are symmetric matrices. Montanari and Sen showed that with probability at least $1 - Ce^{-n/C}$, we have (Theorem 8 equation 47 of [MS16])

$$\max\left\{\frac{1}{n\sqrt{d}}\mathcal{OPT}_k(\pm\boldsymbol{A}_G^{\text{cen}}), \frac{1}{n}\mathcal{OPT}_k(\pm\boldsymbol{B}(\lambda))\right\} \leq C,$$

so in fact, for $\boldsymbol{M} = \boldsymbol{A}_G^{\text{cen}}$ or $\boldsymbol{B}(\lambda)$, with probability at least $1 - Ce^{-n/C}$, we have

$$|\mathcal{SDP}(\boldsymbol{M}) - \mathcal{OPT}_k(\boldsymbol{M})| \leq \frac{C}{k}.$$

Theorem 5.2 (OPT vs Φ, Lemma 3.2 of [MS16])**.** *If $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ satisfies*

$$\|\boldsymbol{M}\|_{\infty\to 2} := \max_{||\boldsymbol{v}||_\infty \leq 1} ||\boldsymbol{M}\boldsymbol{v}||_2 \leq L\sqrt{n},$$

then for any $\beta > 0$, $k \in [n]$, and $\epsilon \in (0, 1]$, there exists a constant C, such that

$$\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \frac{1}{n}\mathcal{OPT}_k(\boldsymbol{M})\right| \leq 2L\epsilon\sqrt{k} + \frac{k}{\beta}\log\frac{C}{\epsilon}.$$

Remark 5.3. We can optimize the bound above over ϵ to obtain that when $\epsilon^* = \sqrt{k}/2\beta L \in (0, 1]$, we have

$$\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \frac{1}{n}\mathcal{OPT}_k(\boldsymbol{M})\right| \leq \frac{k}{\beta}\left(1 + \log\left(\frac{2C\beta L}{\sqrt{k}}\right)\right) = \frac{k}{\beta}\log\frac{C\beta L}{\sqrt{k}}.$$

Remark 5.4. In particular, we can show that (Lemma A.3 of [MS16]) for any graph G with bounded degree, we have $\|\boldsymbol{A}_G^{\text{cen}}\|_{\infty\to 2} \leq Cd\sqrt{n}$, and for any $\lambda \in \mathbb{R}$, we have $\|\boldsymbol{B}(\lambda)\|_{\infty\to 2} \leq (C + |\lambda|)\sqrt{n}$. so the above bound applies to $\boldsymbol{A}_G^{\text{cen}}$ and $\boldsymbol{B}(\lambda)$ with $L = Cd$ and $L = C + |\lambda|$, respectively.

They further showed the following concentration result for Φ.

Theorem 5.3 (Concentration of Φ, Lemma A.1 of [MS16])**.** *If $\boldsymbol{M} \in \mathbb{R}^{n\times n}$ is $\boldsymbol{A}_G^{\text{cen}}$ for some graph G with bounded degree, or $\boldsymbol{B}(\lambda)$ for some $\lambda \in \mathbb{R}$, then for any $\beta > 0$ and $k \leq n$, for all $t > 0$, we have*

$$\mathbb{P}\left(\left|\frac{1}{n}\Phi(\beta, k; \boldsymbol{M}) - \mathbb{E}\left(\frac{1}{n}\Phi(\beta, k; \boldsymbol{M})\right)\right| \geq t\right) \leq Ce^{-nt^2/C}.$$

Therefore, the problem of approximating $\mathcal{SDP}(\boldsymbol{A}_G^{\text{cen}})$ with $\mathcal{SDP}(\boldsymbol{B}(\lambda))$ can be reduced to the problem of interpolating between $\mathbb{E}(\Phi(\beta, k; \boldsymbol{A}_G^{\text{cen}}))$ and $\mathbb{E}(\Phi(\beta, k; \boldsymbol{B}(\lambda)))$. We will defer the details of the interpolation step to the next section, where we examine a more general family of random graph models, the perturbed

SBMs. As it turns out, the interpolation step will be the most central step in extending the algorithm to the new setting.

5.2 Lindeberg Interpolation

The motivation of the interpolation step is the following Lindeberg interpolation lemma. cite

Lemma 5.1 (Lindeberg interpolation)**.** *Let $F : \mathbb{R}^N \to \mathbb{R}$ be a thrice continuously differentiable function, and let $\boldsymbol{X} = (X_1, \ldots, X_N)$ and $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ be two vectors of independent random variables satisfying $\mathbb{E}(X_i) = \mathbb{E}(Z_i)$ and $\mathbb{E}(X_i^2) = \mathbb{E}(Z_i^2)$ for all $i \in [N]$. Then, we have*

$$|\mathbb{E}\left(F(\boldsymbol{X}) - F(\boldsymbol{Z})\right)| \leq \frac{1}{6}\left(\max_{i\in[N]} \|\partial_i^3 F\|_\infty\right)\left(\sum_{i=1}^N \mathbb{E}\left[|X_i|^3 + |Z_i|^3\right]\right).$$

Thus, we construct a matrix $\boldsymbol{D}$, such that $\boldsymbol{D}$ is a Gaussian matrix with first and second moments matching those of $\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$. Then, apply the above lemma to $F(\boldsymbol{M}) := \Phi(\beta, k; \boldsymbol{M})$ with $\boldsymbol{X} = \boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$ and $\boldsymbol{Z} = \boldsymbol{D}$, we can bound the difference between $F(\boldsymbol{A}_G^{\text{cen}}/\sqrt{d})$ and $F(\boldsymbol{D})$.

First, we bound the third moments of the entries of $\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}$. Note that if $X \sim \text{Ber}(p)$, then for $a \in [0,1]$, $\mathbb{E}[|X-a|^3] = (1-p)a^3 + p(1-a)^3 \leq a + p$. Since $\boldsymbol{A}_{G_{ij}}^{\text{cen}} = \boldsymbol{A}_{G_{ij}} - d/n$, and $\boldsymbol{A}_{G_{ij}} \sim \text{Ber}\left(\frac{d}{n}(1+\eta_{ij})\right)$, we have

$$\sum_{i,j\in[n]} \mathbb{E}\left|\frac{1}{\sqrt{d}}\boldsymbol{A}_{G_{ij}}^{\text{cen}}\right|^3 \leq \frac{1}{d^{3/2}}\sum_{i,j\in[n]}\frac{d}{n}(2+\eta_{ij}) \leq \frac{2n}{\sqrt{d}} + \frac{1}{n\sqrt{d}}\sum_{i,j\in[n]} h_{ij} < \frac{3n}{\sqrt{d}}.$$

finish the argument above

Next, we bound the third moments of the entries of $\boldsymbol{D}$. Note that if $X = a + bZ$, where $Z \sim \mathcal{N}(0,1)$, then $\mathbb{E}[|X|^3] \leq 4(|a|^3 + |b|^3)$. Since $\boldsymbol{D}_{ij} = -p_{ij}/\sqrt{d} + \sqrt{p_{ij}(1-p_{ij})/d}Z_{ij}$, where $Z_{ij} \sim \mathcal{N}(0,1)$, we have

$$\begin{aligned}
\sum_{i,j\in[n]} \mathbb{E}\left|\boldsymbol{D}_{ij}\right|^3 &\leq 4\sum_{i,j\in[n]}\left(\left(\frac{p_{ij}}{\sqrt{d}}\right)^3 + \left(\frac{p_{ij}(1-p_{ij})}{d}\right)^{3/2}\right) \\
&\leq \sum_{i,j\in[n]}\left(\frac{d^{3/2}}{n^3}(1+\eta_{ij})^3 + \frac{1}{n^{3/2}}(1+\eta_{ij})^{3/2}\right) \\
&\leq \frac{8d^{3/2}}{n} + 2\sqrt{2n} < \frac{n}{\sqrt{d}},
\end{aligned}$$

where we used the fact that $|\eta_{ij}| \leq 1$ and $d \ll n$. clean up argument

Finally, Montanari and Sen proved that $\|\partial_{ij}^3 F\|_\infty \leq 6\beta^2$ for all $i, j \in [n]$ (line 112 of [MS16]). Combining, we obtain

$$\left|\mathbb{E}\left(F(\boldsymbol{A}_G^{\text{cen}}/\sqrt{d}) - F(\boldsymbol{D})\right)\right| \leq \frac{Cn\beta^2}{\sqrt{d}}$$

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5.3 Coupling of Gaussian Matrices

For simplicity, we write $\boldsymbol{B}(\lambda) = \boldsymbol{B}$. To bound the difference between $F(\boldsymbol{D})$ and $F(\boldsymbol{B}(\lambda))$, we first notice that, for all matrices $\boldsymbol{M}, \boldsymbol{M}' \in \mathbb{R}^{n\times n}$, we have

$$\begin{aligned}
F(\boldsymbol{M}) &= \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n \boldsymbol{M}_{ij}\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n \boldsymbol{M}'_{ij}\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle + \beta\sum_{i,j=1}^n (\boldsymbol{M}_{ij}-\boldsymbol{M}'_{ij})\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\sum_{i,j=1}^n (\boldsymbol{M}_{ij}-\boldsymbol{M}'_{ij})\langle\boldsymbol{\sigma}_i,\boldsymbol{\sigma}_j\rangle\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\operatorname{Tr}(\boldsymbol{\Sigma}(\boldsymbol{M}-\boldsymbol{M}')\boldsymbol{\Sigma}^\top)\right) d\nu(\boldsymbol{\sigma})\right)\\
&\le F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(\beta\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}\operatorname{Tr}(\boldsymbol{\Sigma}^\top\boldsymbol{\Sigma})\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + \frac{1}{\beta}\log\left(\int_{(\mathbb{S}^{k-1})^n} \exp\left(n\beta\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}\right) d\nu(\boldsymbol{\sigma})\right)\\
&= F(\boldsymbol{M}') + n\|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}.
\end{aligned}$$

where we wrote $\boldsymbol{\Sigma} := (\boldsymbol{\sigma}_1,\dots,\boldsymbol{\sigma}_n) \in (\mathbb{S}^{k-1})^n$, and used the fact that for $\boldsymbol{A}$ symmetric and $\boldsymbol{B} \succeq 0$, since $\boldsymbol{A} \preceq \|\boldsymbol{A}\|_{\mathrm{op}}\boldsymbol{I}$, we have

$$\operatorname{Tr}(\boldsymbol{AB}) \le \operatorname{Tr}(\|\boldsymbol{A}\|_{\mathrm{op}}\boldsymbol{IB}) \le \|\boldsymbol{A}\|_{\mathrm{op}}\operatorname{Tr}(\boldsymbol{B}).$$

Hence,

$$\left|\frac{1}{n}F(\boldsymbol{M}) - \frac{1}{n}F(\boldsymbol{M}')\right| \le \|\boldsymbol{M}-\boldsymbol{M}'\|_{\mathrm{op}}.$$

Therefore,

$$\begin{aligned}
\frac{1}{n}\left|\mathbb{E}F(\boldsymbol{D}) - \mathbb{E}F(\boldsymbol{B})\right| &\le \frac{1}{n}\mathbb{E}\left|F(\boldsymbol{D}) - F(\boldsymbol{B})\right|\\
&\le \mathbb{E}\|\boldsymbol{D}-\boldsymbol{B}\|_{\mathrm{op}}.
\end{aligned}$$

Thus, we want to find a bound on $\mathbb{E}\|\boldsymbol{D}-\boldsymbol{B}\|_{\mathrm{op}}$. To do this, we will use a coupling argument. Writing $V_{ij} = np_{ij}(1-p_{ij})/d$, note that

$$V_{ij} = (1+\eta_{ij})\left(1 - \frac{d}{n}(1+\eta_{ij})\right),$$

$$\boldsymbol{D}_{ij} \sim \mathcal{N}\left(\frac{\lambda}{n}x_ix_j + \frac{\sqrt{d}}{n}h_{ij},\ \frac{1}{n}V_{ij}\right),$$

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$$\boldsymbol{B}_{ij} \sim \mathcal{N}\left(\frac{\lambda}{n} x_i x_j,\ \frac{1}{n}\right).$$

Thus, we can couple $\boldsymbol{D}$ and $\boldsymbol{B}$ such that

diagonal

$$\boldsymbol{D} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \frac{\sqrt{d}}{n}\boldsymbol{H} + \boldsymbol{Z_0} + \boldsymbol{Z_D},$$

$$\boldsymbol{B} = \frac{\lambda}{n}\boldsymbol{x}\boldsymbol{x}^\top + \boldsymbol{Z_0} + \boldsymbol{Z_B},$$

where $\boldsymbol{Z_0}, \boldsymbol{Z_D}, \boldsymbol{Z_B}$ are Gaussian matrices with mean-zero, independent entries. Writing $V_- = \min_{i,j} V_{ij}$, the variances of the entries are given by

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{0},ij}) = \begin{cases} \frac{1}{n}V_- & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{D},ij}) = \begin{cases} \frac{1}{n}(V_{ij} - V_-) & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{B},ij}) = \begin{cases} \frac{1}{n}(1 - V_-) & \text{if } i \neq j, \\ 0 & \text{if } i = j, \end{cases}$$

Under this coupling, by the triangle inequality, we have

$$\mathbb{E}\|\boldsymbol{D} - \boldsymbol{B}\|_{\mathrm{op}} = \mathbb{E}\left\|\frac{\sqrt{d}}{n}\boldsymbol{H} + \boldsymbol{Z_D} - \boldsymbol{Z_B}\right\|_{\mathrm{op}} \leq \frac{\sqrt{d}}{n}\mathbb{E}\|\boldsymbol{H}\|_{\mathrm{op}} + \mathbb{E}\|\boldsymbol{Z_D}\|_{\mathrm{op}} + \mathbb{E}\|\boldsymbol{Z_B}\|_{\mathrm{op}}.$$

We then use the following result for operator norm of centered Gaussian matrices.

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theorem statement gives op bound of H

cite, 1408.6185

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Theorem 5.4 (Operator norm of Gaussian matrices)**.** *Let $\boldsymbol{X}$ be an $n \times n$ symmetric matrix with $\boldsymbol{X}_{ij} = g_{ij} b_{ij}$ for $i \geq j$, where g_{ij} are independent standard Gaussian variables, and b_{ij} are given scalars. Then, writing*

$$\sigma^*_{col} = \max_{j \in [n]} \sqrt{\sum_{i=1}^{n} b_{ij}^2}, \quad \sigma^*_{elt} = \max_{i,j \in [n]} |b_{ij}|,$$

for every $s \in (0, 1/2]$, we have

$$\mathbb{E}\|\boldsymbol{X}\| \leq (1+s)\left(2\sigma^*_{row} + \frac{6}{\sqrt{\log(1+s)}}\sigma^*_{elt}\sqrt{\log n}\right)$$

In our case, writing $V_+ = \max_{i,j} V_{ij}$, for all $i, j \in [n]$, we have

$$\mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{D},ij}),\ \mathrm{Var}(\boldsymbol{Z}_{\boldsymbol{B},ij}) \leq \frac{1}{n}(V_+ - V_-).$$

Thus, writing $\delta V := V_+ - V_-$, we have $\sigma^*_{\text{col}} \le \sqrt{\delta V}$, $\sigma^*_{\text{elt}} \le \sqrt{\delta V/n}$. Applying the above theorem,

$$\mathbb{E}\|\boldsymbol{Z_D}\|,\ \mathbb{E}\|\boldsymbol{Z_B}\| \le (1+s)\left(2 + \frac{6}{\sqrt{\log(1+s)}}\sqrt{\frac{\log n}{n}}\right)\sqrt{\delta V} \le C\sqrt{\delta V},$$

when n is sufficiently large.

Recall that $V_{ij} = np_{ij}(1-p_{ij})/d$, and $p_{ij} \ll 1/2$, so V_{ij} is increasing in p_{ij}, and thus increasing in η_{ij}. Since $|\eta_{ij}| \le \epsilon/2 + \|h\|_\infty$, we have

$$\begin{aligned}
\delta V = V_+ - V_- &\le \left(1 + \frac{\epsilon}{2} + \|h\|_\infty\right)\left(1 - \frac{d}{n}\left(1 + \frac{\epsilon}{2} + \|h\|_\infty\right)\right) \\
&\quad - \left(1 - \frac{\epsilon}{2} - \|h\|_\infty\right)\left(1 - \frac{d}{n}\left(1 - \frac{\epsilon}{2} - \|h\|_\infty\right)\right) \\
&\le \left(1 - \frac{d}{n}\right)(\epsilon + 2\|h\|_\infty) \\
&< \epsilon + 2\|h\|_\infty < Cd^{-2\alpha}.
\end{aligned}$$

Therefore, we have

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$$\begin{aligned}
\mathbb{E}\|\boldsymbol{D} - \boldsymbol{B}\|_{\text{op}} &\le \frac{\sqrt{d}}{n}\mathbb{E}\|\boldsymbol{H}\|_{\text{op}} + \mathbb{E}\|\boldsymbol{Z_D}\|_{\text{op}} + \mathbb{E}\|\boldsymbol{Z_B}\|_{\text{op}} \\
&\le \frac{\sqrt{d}}{n}\left(Cd^{-1/2-\alpha}\right) + C\sqrt{\delta V} \\
&= Cd^{-\alpha}.
\end{aligned}$$

Combining with the bound from the Lindeberg interpolation step, we have

$$\frac{1}{n}\left|\mathbb{E}F\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathbb{E}F(\boldsymbol{B})\right| \le C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right).$$

5.4 Combining all the pieces

With the above bound between $\mathbb{E}F$ values, we first use the concentration of Φ from Theorem 5.3 to show that taking $t = O(d^{-1/2})$, with probability at least $1 - Ce^{-n/C}$, we have

$$\frac{1}{n}\left|\Phi\left(\beta, k; \frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \Phi(\beta, k; \boldsymbol{B})\right| \le C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right).$$

Then, Theorem 5.2 allows us to relate Φ and $\mathcal{OPT}_k$, which gives

$$\begin{aligned}
\frac{1}{n}\left|\mathcal{OPT}_k\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{OPT}_k(\boldsymbol{B})\right| &\le \frac{k}{\beta}\log\frac{C\beta d}{\sqrt{k}} + \frac{k}{\beta}\log\frac{C\beta\lambda}{\sqrt{k}} + C\left(\frac{\beta^2}{\sqrt{d}} + d^{-\alpha}\right) \\
&\le \frac{k}{\beta}\log\frac{C\beta d\lambda}{k} + C\frac{\beta^2}{\sqrt{d}} + Cd^{-\alpha}.
\end{aligned}$$

Note that this holds for all $\beta \in \mathbb{R}$, so taking $\beta = d^{1/6}k^{1/3}$, we have

$$\begin{aligned}\frac{1}{n}\left|\mathcal{OPT}_k\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{OPT}_k(\boldsymbol{B})\right| &\leq Cd^{-1/6}k^{2/3}\log(Cd^{7/6}k^{-2/3}\lambda) + Cd^{-\alpha} \\ &\leq Cd^{-1/6}k^{2/3}\log(Cd^{7/6}\lambda) + Cd^{-\alpha},\end{aligned}$$

where we dropped the $k^{-2/3}$ term in the log since k is a positive integer.

Finally, Theorem 5.1 links $\mathcal{OPT}_k$ and $\mathcal{SDP}$, which gives

$$\frac{1}{n}\left|\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{SDP}(\boldsymbol{B})\right| \leq \frac{C}{k} + Cd^{-1/6}k^{2/3}\log(Cd^{7/6}\lambda) + Cd^{-\alpha}.$$

Taking $k = d^{1/10}$, we have

$$\frac{1}{n}\left|\mathcal{SDP}\left(\frac{1}{\sqrt{d}}\boldsymbol{A}_G^{\text{cen}}\right) - \mathcal{SDP}(\boldsymbol{B})\right| \leq C\frac{\log(Cd^{7/6}\lambda)}{d^{1/10}} + Cd^{-\alpha}.$$

Crucially, the above bound goes to zero as $d \to \infty$, which allows us to conclude that the same algorithm still works for the perturbed SBM, proving Theorem 4.1. ■

6 Extension to node robustness

References

[MS16] Montanari, A. and Sen, S. (2015) *Semidefinite Programs on Sparse Random Graphs and their Application to Community Detection*, Proceedings of the forty-eighth annual ACM symposium on Theory of Computing, 2016, pp. 814–827.

[1] Author (Year) *Title*, Publisher